Technical Architecture & Spatial Indexing Framework

This document details the methodological framework, mathematical formulations, foundational sources, and Python code implementation for normalizing diverse urban environmental datasets into a unified 0–100 Green Score index across spatial units (e.g., Manhattan Community Districts or Census Tracts).

1. Executive Summary & Core Objective

Urban environmental datasets span distinct measurement scales, physical units, and directional relationships (e.g., higher tree canopy is positive, whereas higher air pollution is negative). To synthesize these inputs into an aggregate Green Score, every raw feature is transformed into a standardized 0 to 100 sub-score where:

- **100** represents optimal environmental performance / minimal vulnerability.
- **0** represents poor environmental quality / maximum vulnerability.

2. Methodology, Formulas, & Web Citation Sources

2.1 Heat Vulnerability Index (HVI)

- **Raw Format:** Ordinal scale from 1 (lowest risk) to 5 (highest risk).
- **Method:** Inverted Linear Scale Normalization.
- **Formula:** Heat_Score = (1 - (HVI - 1) / 4) * 100
- **Reasoning:** Inverts vulnerability metrics so that lower risk maps to 100. Step increments of 1 HVI unit correspond to 25-point sub-score adjustments (1 = 100, 2 = 75, 3 = 50, 4 = 25, 5 = 0).

2.2 Urban Tree Canopy Cover

- **Raw Format:** Continuous percentage (0% to 100%).
- **Method:** Target-Capped Linear Scaling.
- **Formula:** Canopy_Score = min(100, (Canopy_Pct / 50.0) * 100)
- **Primary Web Sources & Citations:**
    - NYC Comptroller: Funding & Managing NYC's Urban Forest and Tree Canopy
    - USFS i-Tree Canopy Technical Manual & Guidelines
- **Reasoning:** While citywide initiatives establish a 30% baseline goal, an upper bound ceiling of 50% benchmarks top-tier residential and park-adjacent census tracts in

Manhattan (e.g., Central Park perimeter), establishing a full 0–100 spread without penalizing dense urban blocks.

2.3 Air Quality (Fine Particulate Matter - PM2.5)

- **Raw Format:** Continuous concentration in µg/m³.
- **Method:** Inverted Min-Max Feature Scaling.
- **Formula:** Air_Score = (1 - (PM2.5 - PM2.5_min) / (PM2.5_max - PM2.5_min)) * 100
- **Primary Web Sources & Citations:**
  - NYC Environment & Health Data Portal – Community Air Survey (NYCCAS)
  - EPA National Ambient Air Quality Standards (NAAQS) Table
- **Reasoning:** Reverses pollution concentrations so cleaner air receives higher scores. Scaling anchors dynamically to NYCCAS empirical survey boundaries (~5.5 to ~14.0+ µg/m³) and EPA NAAQS annual standards (9.0 µg/m³).

2.4 Spatial Multi-Criteria Decision Analysis (SMCDA) Framework

- **Primary Web Sources & Citations:**
  - Analytic Hierarchy Process (AHP) Technical Paper (T.L. Saaty)
  - CHARIM Guidebook: Spatial Multi-Criteria Evaluation (SMCE)
  - NSF Repository: Prediction of Building Energy Star Score (NYC Open Data)
- **Reasoning:** Formulates the mathematical foundation for GIS-based min-max feature scaling and linear weighted aggregate scoring.

3. Mathematical Formulation Summary Table

Indicator	Raw Unit	Relationship	Transformation Formula	Key Source / Benchmark
Heat Index	1 – 5 Scale	Inverse (Lower is better)	(1 - (HVI - 1) / 4) * 100	NYC DOHMH HVI Scale
Tree Canopy	Percentage (%)	Direct (Higher is better)	min(100, (Canopy / 50.0) * 100)	NYC Forest Agenda / USFS i-Tree
Air Quality	PM2.5 (µg/m³)	Inverse (Lower is better)	(1 - (PM2.5 - Min) / (Max - Min)) * 100	NYCCAS Survey / EPA NAAQS
Energy Efficiency	Site EUI (kBtu/ft²)	Inverse (Stratified by prop…	(1 - (EUI - p5) / (p95 - p5)) * 100	ACEEE Study / NYC LL84 Data